Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Lambert, James L Ph.D.		
Project Title:	Microbial Water Monitoring using a Multiplexed Lateral Flow Assay		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-4181
Organization Name:	NASA Jet Propulsion Laboratory		
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PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/2000	**End Date:**	09/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fisher, Anita (Jet Propulsion Laboratory)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Multi-agent immunoassays for monitoring the integrity of the food and water will be essential for maintaining astronaut health during long duration space flight. Similar technology may also be applied for monitoring the dynamics of the ecology within bioreactors or for clinical testing of astronauts during flight. Ideally, the design of these devices should be such that a minimum of crew interaction is required for their maintenance or use. Our laboratory is developing two types of multiplexed immunoassays for water monitoring that incorporate quantum dots for measuring the concentration of several types of pathogen in a single test. Both immunoassay tests are intended to be used in conjunction with the water test kit (WTK) currently used on the ISS. Lateral flow assays (LFA) and flow-through assays (FTA) are being developed that provides single-use testing for the presence of analytes within a fluid sample. LFA's are strip tests that are typically read visually and require no active fluidics system since they rely only on capillary action for fluid flow. FTA may be constructed on microbial filters to allow in-place testing of microbes within the current WTK system used		

	on the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Quantum dots provide a rich pallet of colors that may be used in an inexpensive, easy-to-use test that can be designed to screen for a variety of cancers, food or water pathogens in a single test strip costing a few dollars.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)